Engineering Portfolio

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Electromechanical Systems • CAD Design • Embedded Electronics • Product Development

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Project :

10m Electronic Target System & 10m Running Target Machine

2023 – 2026



PROJECT OVERVIEW

10m Electronic Target System

Problem

Traditional target systems are expensive and difficult to obtain locally.

Objective

Develop a commercially viable electronic target system capable of supporting shooting sport competitions with low cost and make it affordable for local purchase

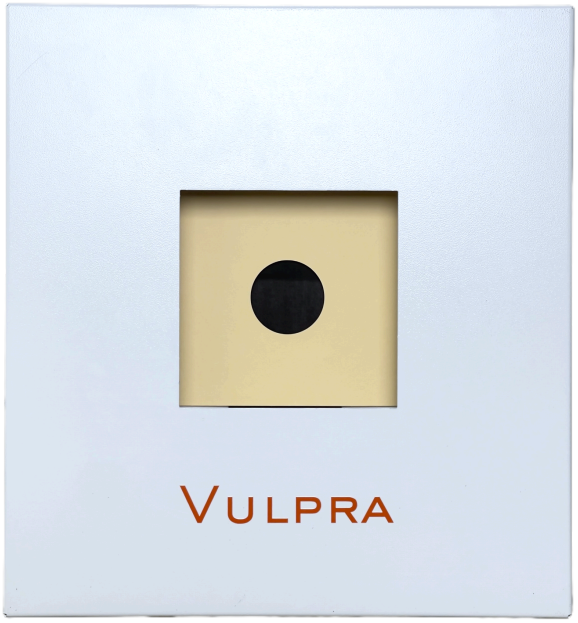
Project Scope

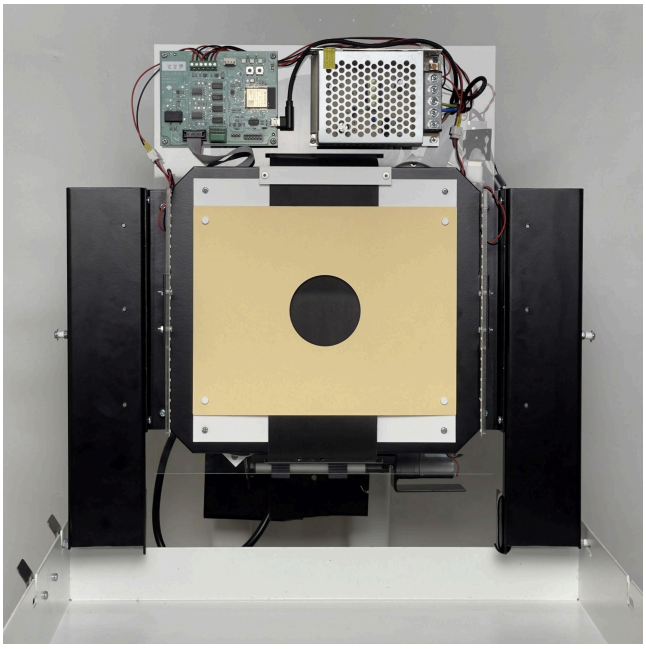
- Mechanical Design
- Electronics Integration
- Embedded Systems
- Assembly Process
- Manufacturing Preparation
- Product Standardization

My Role

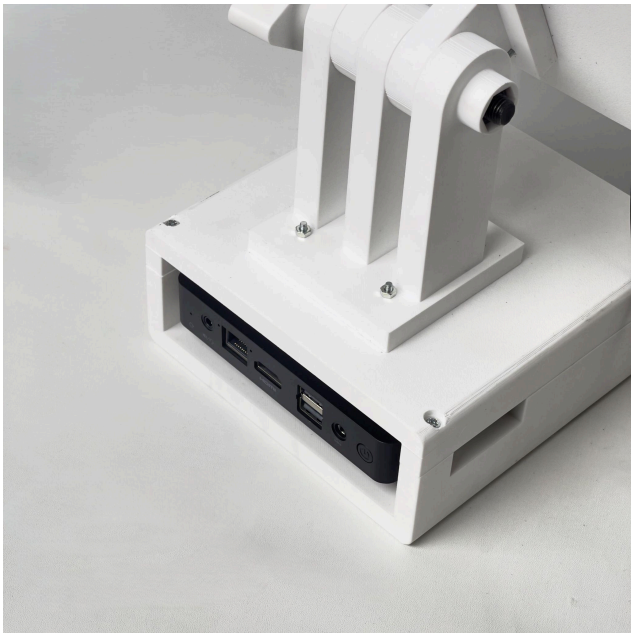
Responsibilities:

- System architecture
- Mechanical design
- Electronics integration
- Logic design
- Assembly
- Testing
- Documentation
- Budget planning
- Customer communication


VULPRA

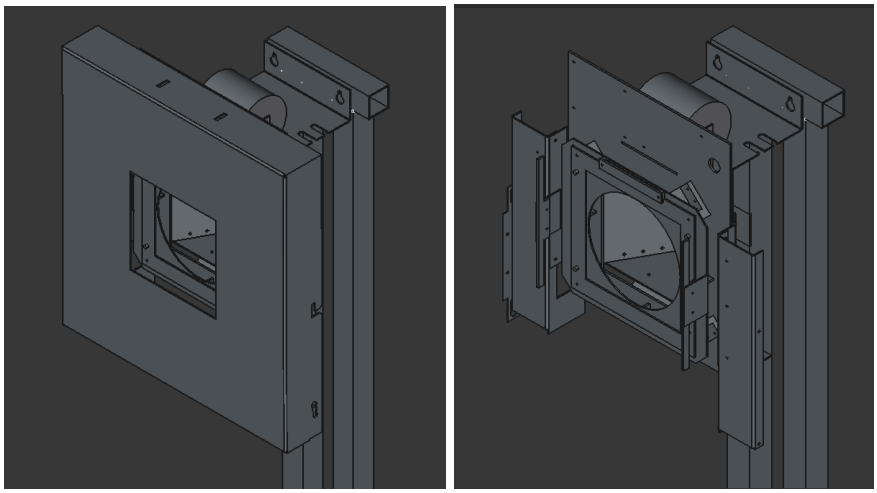



VULPRA

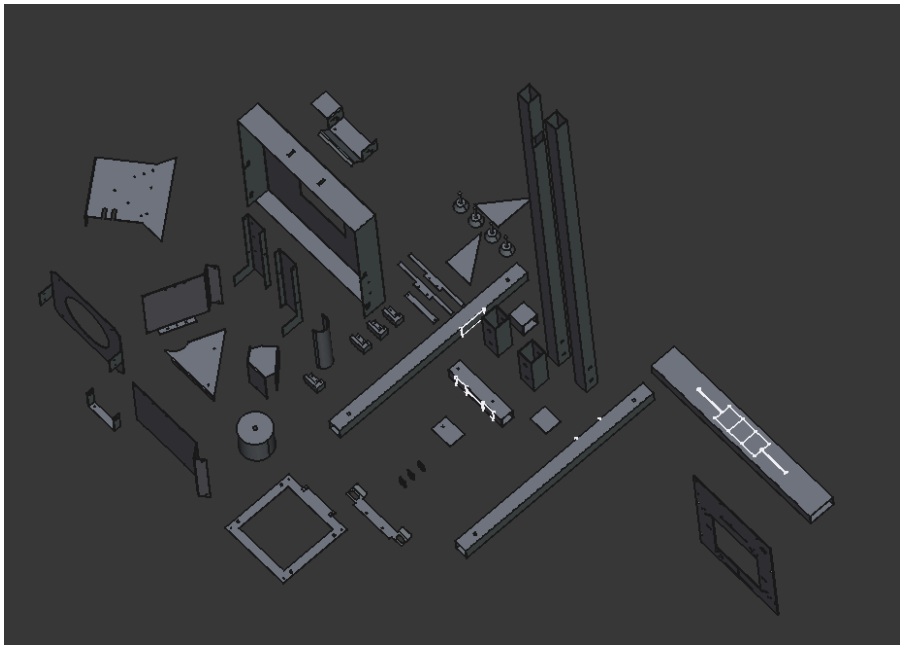


CAD DESIGN

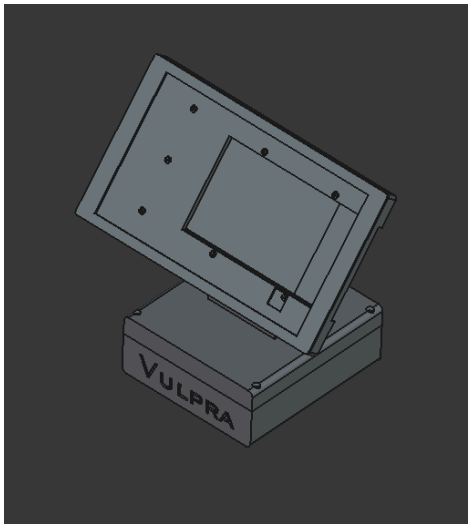
Mechanical Design

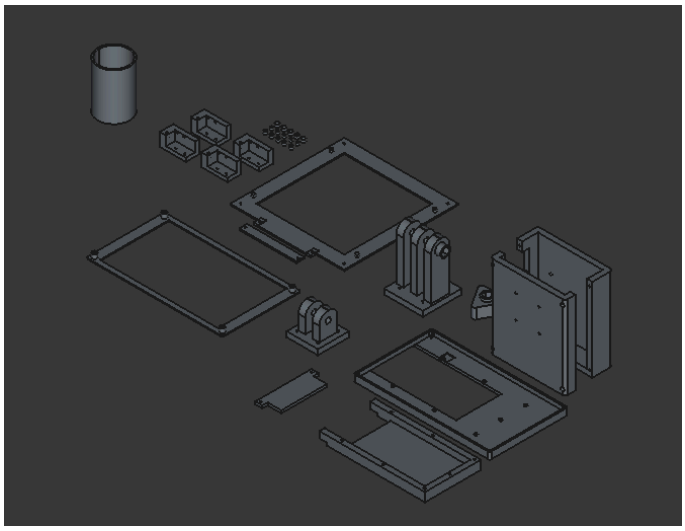


This is an overview of the whole mechanical design



This is roughly all the part that i designed




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This is the rough overview of the monitor design

Design Considerations

- Ease of manufacturing
- Cost optimization
- Reliability
- Ease of maintenance

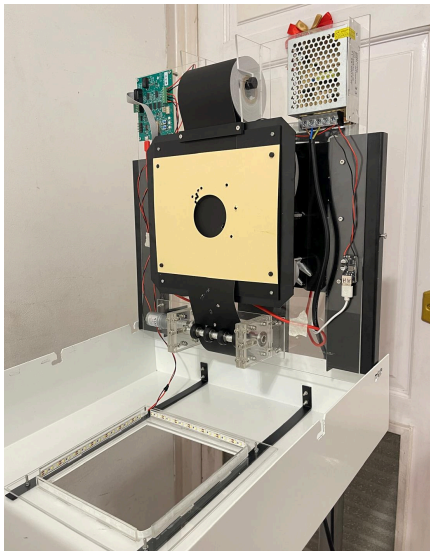
Development Process

Challenge

Create a reliable target system and make it design efficient

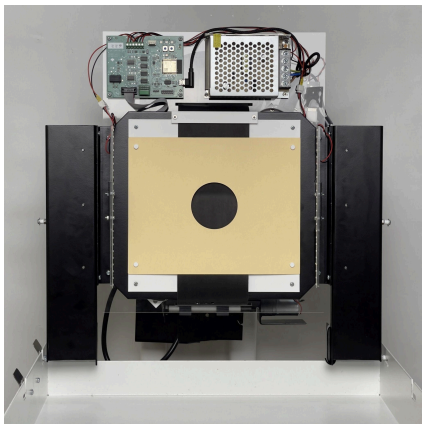
Development

Version 1



Problem : The design is not efficient enough and too complicated. That's why the machine weight and the amount of parts is higher. It is also harder to assemble.

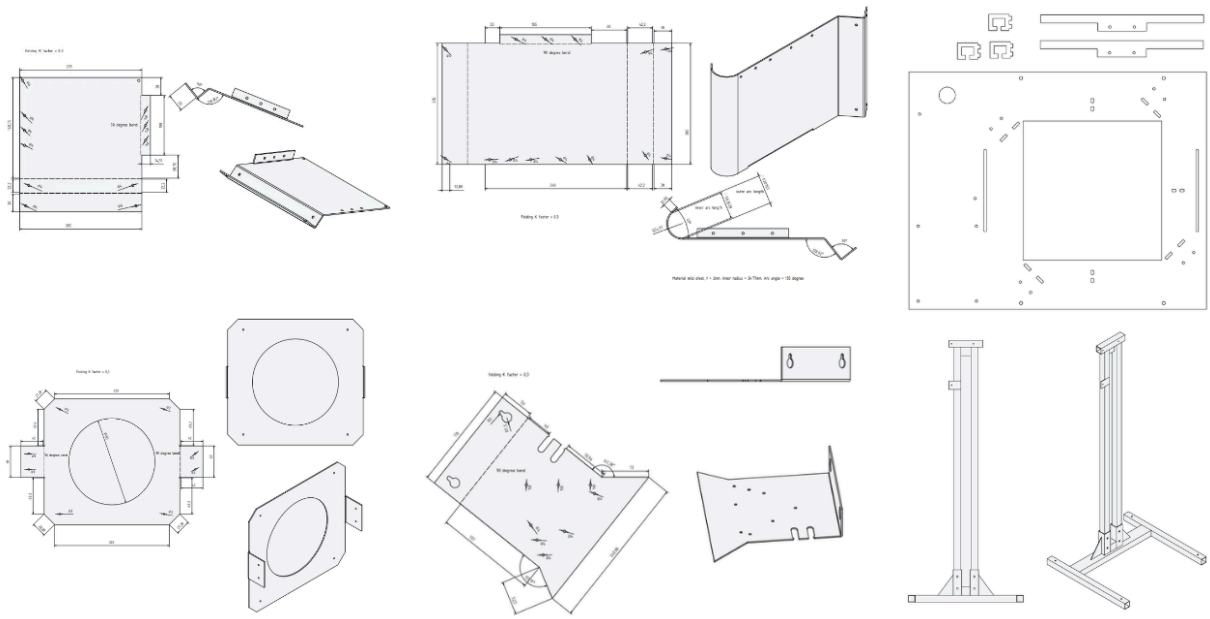
Version 2 (Final Product)



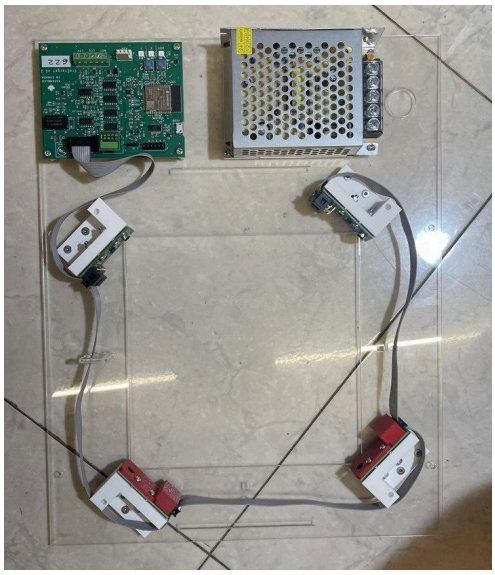
With some improvement and design efficiency, I manage to reduce the parts count, the weight of the machine, and the simplicity of the assembly without reducing the overall function.

Manufacturing & Documentation

Manufacturing design



These are some overview of the technical design for manufacturing process, the rest of the file are in DXF and PDF form and can be viewed on my github.



This is the overview of the internal component before fully assembled.

10m Running Target

The project documentation is in this google drive link :
Running target portfolio

Thank You

For further information, CAD files and technical documentation:

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Portfolio Repository: https://github.com/redfoxxxi/ET_MASTER_CODE